Natural Language Processing Case Study

ResolvAI: An Intelligent Customer Support Chatbot using Retrieval-Augmented Generation (RAG)

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Product Name: ResolvAI — an intelligent, retrieval-augmented customer support chatbot for the e-commerce domain.
Slogan: "Support that knows, not guesses."

1 Introduction

Customer support is the front door of every digital business. Every day, e-commerce platforms receive thousands of queries about order status, returns, refunds, payments, shipping, and account issues. The majority of these questions are repetitive, and their answers already exist inside the organisation — in FAQ pages, policy documents, product manuals, and previously resolved tickets. Yet customers still wait in long queues for human agents, and support teams spend most of their time answering the same questions again and again.

ResolvAI is a smart customer support chatbot which operates on the Retrieval-Augmented Generation (RAG) framework, an NLP architecture which combines the understanding abilities of the Large Language Models (LLMs) with the accuracy of information from the organization's knowledge base. While ResolvAI does not generate answers based on its own memory but first finds the most relevant information from the documents of the company, and then formulates the answer using the information provided in them. In case no answer is found in the knowledge base, the bot admits it and passes the control to human agents rather than trying to make up an answer. ResolvAI uses the three key NLP concepts which we have defined for this case study – intent classifi- cation, dialogue systems and the Large Language Models. The fine-tuned DistilBERT classifier identifies the intent of the customer (order tracking, refund request, cancellation, etc.), the dialogue management layer rewrites the multi-turn follow-up questions into stand-alone queries, and the LLM produces the final grounded response. These are surrounded by classical NLP tools – text preprocessing and nor- malization, Named Entity Recognition for masking PII, BM25 and Sentence-BERT for representing texts, and principled evaluation.

This case study presents the problem ResolvAI addresses, the research gaps it targets, a survey of related commercial products, its mapping to the UN Sustainable Development Goals, the dataset used to build and evaluate it, and the novelty of the proposed system.

2 Problem Statement

2.1 What is the problem / pain point being solved?

Organizations cannot provide immediate, precise, and trustworthy responses to customers' questions in mass volume. It is very inefficient for businesses to rely solely on humans since customers have to wait minutes or even hours for an answer that can be easily found in documentation; and agents will become demotivated after answering hundreds of tier-1 tickets that require the same answer.

The current automation is ineffective for the following reasons:

- **Keyword matching chatbots** are not flexible enough because they are based on prewritten scripts and cannot operate when a customer asks a question in a different way (e.g., "my package is not here" and "track order").

- **LLM only chatbots** understand natural language very well but have no access to the organization's policy, thus they hallucinate. It means that they make up answers to the customer's questions.

In terms of customer support, the worst-case scenario is an incorrect answer that is provided with full confidence (e.g., a promise of providing a refund when there is no such policy).

2.2 What is the proposed solution?

ResolvAI is a RAG (Retrieval-Augmented Generation) chatbot. All organizational knowledge – FAQs, user guides, terms and conditions concerning returns and requesting refund on products, as well as closed support tickets – is preprocessed, chunked, semantically vectorized, and stored in the vector database.

Upon receiving customer queries, the most relevant chunks are selected through hybrid search (the combination of semantic similarity and keyword-based search), and then used as input for language model which generates the answer, based exclusively on the retrieved documents. Every answer is accompanied by a citation to the original source. In case if the retrieval confidence score is below the threshold, the bot does not attempt to make assumptions and passes the query to a human operator along with the conversation history.

2.3 How does the solution work, in layman terms?

- A standard AI chatbot sits for a closed book examination; it replies from memory, and whenever it does not know, it lies.
- ResolvAI sits for an open book examination. It opens the exact pages in the firm's internal documents that have information about the client's query, reads those pages, and then responds using those pages – even giving a reference to the "page" it has read.
- In case there is no information on the subject in the book at all, it lifts its hand and asks the teacher (human agent) to help.

Essentially, ResolvAI marries a very quick librarian with an excellent writer: the former locates the necessary documents, while the latter converts them into a helpful response without the liberty of fabricating any facts.

3 Motivation

- **Scale of repetition:** Around 70–80% of customer support queries are repetitive tier-1 questions whose answers already exist in documentation; automating them frees human agents for genuinely complex cases.
- **Customer expectations:** Modern customers expect 24×7 instant responses; long wait times directly translate into cart abandonment, churn, and negative reviews.
- **Cost of hallucination:** Hallucination in LLM chatbots is not a cosmetic flaw — companies have faced real financial and legal consequences when their bots gave customers incorrect policy information. Grounded, cited answers are a trust requirement, not a luxury.
- **Accessibility gap:** Enterprise AI support products are expensive and proprietary; small and medium businesses are priced out. An open, reproducible RAG pipeline democratises quality support automation.
- **Academic relevance:** RAG sits at the intersection of information retrieval, semantic search, and text generation — the most active research areas in NLP today — making this an academically rich case study with direct industrial relevance.

4 Research Questions / Gaps to be Addressed

- **RQ1.** How can a support chatbot reliably detect that the answer to a query if the data is not present in its knowledge base, and abstain or escalate instead of hallucinating? *(Gap: existing bots lack confidence-aware fallback.)*
- **RQ2.** Does hybrid retrieval (dense semantic embeddings + sparse keyword search such as BM25) outperform either method alone on jargon-heavy, misspelled, and informally phrased customer queries? *(Gap: purely semantic retrieval misses exact product codes and error names; purely lexical retrieval misses paraphrases.)*
- **RQ3.** How can multi-turn conversational context ("what about the second option?", "and for international orders?") be incorporated into retrieval so that follow-up questions are resolved correctly? *(Gap: most RAG systems treat every query independently.)*
- **RQ4.** What chunking strategy (size, overlap, structure-awareness) yields the best retrieval precision for heterogeneous support documents such as FAQs, policies, and ticket logs? *(Gap: chunking is usually chosen arbitrarily, yet it strongly affects answer quality.)*
- **RQ5.** How faithfully can generated answers be automatically evaluated for groundedness using reference-free metrics rather than manual inspection alone? *(Gap: evaluation of RAG systems remains ad-hoc in practice.)*

5 Related Products and Their Details

The table below surveys leading commercial AI customer-support products and the limitations that motivate ResolvAI.

Product	URL	Key Features	Limitations
Intercom Fin	`intercom.com/fin`	GPT-4 based agent trained on help-centre content; multi-channel; resolves tickets end-to-end.	Per-resolution pricing becomes very costly at scale; proprietary black box; limited transparency on why an answer was produced.
Zendesk AI	`zendesk.com`	Intent detection, macro suggestions, agent-assist, knowledge-base answer bot integrated into ticketing suite.	Locked into the Zendesk ecosystem; shallow generative grounding; customisation of retrieval behaviour is limited.
Ada	`ada.cx`	No-code automation builder; multilingual; integrates with CRMs and order systems.	Answer quality depends heavily on manually curated flows; weak on unseen phrasing; enterprise pricing.
Tidio Lyro	`tidio.com`	Lightweight AI answer bot for small businesses; quick FAQ ingestion; live-chat handoff.	Small knowledge capacity; struggles with long policy documents and multi-turn context; no source citation.
Forethought	`forethought.ai`	"Agatha" engine for intent triage, retrieval from past tickets, and agent-assist suggestions.	Focused on agent assistance rather than fully grounded customer-facing answers; opaque confidence handling.

Common limitations across these products: high proprietary cost, lack of source citation and explainability, weak or absent "I don't know" behaviour on out-of-knowledge-base queries, and English-centric handling of informal or code-mixed customer language. ResolvAI is designed explicitly around these four gaps.

6 SDG Mapping

6.1 SDGs Identified

- **SDG 9:** Industry, Innovation and Infrastructure
- **SDG 8:** Decent Work and Economic Growth

6.2 How the Case Study Relates to the SDGs

- **SDG 9 (Target 9.b – support domestic technology development and innovation):** ResolvAI builds resilient, intelligent digital infrastructure for customer service. By using open-source models and a reproducible RAG pipeline, it upgrades the technological capability of businesses — particularly small and medium enterprises — without dependence on expensive proprietary platforms.
- **SDG 8 (Target 8.2 – higher productivity through technological upgrading and innovation):** Automating repetitive tier-1 queries raises the productivity of support teams and improves the quality of work for human agents, who shift from monotonous repetition to complex, higher-value problem solving. Affordable automation also lets small businesses offer 24×7 support they could not otherwise staff, supporting their growth and competitiveness.

7 Dataset Details (IEEE Dataport)

7.1 Dataset Description

The primary dataset for this case study is a customer-support conversational corpus of the type hosted on IEEE Dataport (following the sample shared for reference), supplemented by the widely used Bitext Customer Support LLM Chatbot Training Dataset.

The corpus includes customer questions paired with responses from agents and labeled with the intent category (order tracking, refund request, cancellation, payment problems, account management etc.) along with the higher level domain, which is characteristic for e-commerce support. The dataset includes the actual query of the customer — in natural language with actual spelling mistakes and abbreviation — alongside the text with the solution.

The whole combined corpus is comprised of tens of thousands of question-response pairs for over 25 intents serving both (a) as a source of the knowledge base and (b) as question-answer pairs for evaluation of retrieval accuracy and answer fidelity. The policy texts and FAQ pages are further processed and included in the knowledge base to provide documents from which the answers should be found, making the task more real-life like.

7.2 Specific Challenges Addressed in the Dataset

- **Noise and text cleaning:** removal of URLs, HTML artefacts, signatures, and boilerplate; normalisation of casing, elongated words ("pleaseee"), and emojis; spelling correction for genuine customer typos while preserving product codes and order IDs.
- **PII anonymisation:** customer messages contain names, e-mail addresses, phone numbers, and order numbers; these are detected with NER/regex patterns and masked with placeholder tokens before any indexing or training.
- **Duplication:** near-identical tickets and templated agent replies are deduplicated using cosine similarity on embeddings to prevent the retriever from being flooded by redundant passages.
- **Class imbalance:** some intents (order tracking) are heavily represented while others (warranty claims) are rare; stratified sampling is used when constructing evaluation splits so that minority intents are fairly assessed.
- **Chunking of long documents:** long policy documents must be split into retrieval units; the case study treats chunk size and overlap as an experimental variable (RQ4) rather than a fixed constant.
- **Language variety:** informal Indian-English and code-mixed phrasing present in real queries is preserved (not filtered out) so that robustness to such language can be measured explicitly.

8 Novelty of the Product

RAG itself is an established architecture; the novelty of ResolvAI lies in the specific combination of mechanisms engineered for trustworthy customer support:

1. **Hybrid retrieval:** every query is answered using a fusion of dense semantic retrieval and sparse BM25 keyword retrieval, so both paraphrased questions and exact product/error codes are matched — a weakness of single-method commercial bots.

2. **Confidence-aware abstention and human escalation:** the system computes a retrieval-confidence score for each query; below a learned threshold it refuses to answer and escalates to a human agent with full context, converting hallucination risk into a graceful handoff.

3. **Source-cited answers:** every generated answer cites the exact source passage(s) it was grounded in, giving customers and auditors verifiable transparency that mainstream products do not provide.

4. **Conversational query rewriting:** follow-up questions are rewritten into standalone queries using conversation history before retrieval, addressing the multi-turn context gap (RQ3).

5. **Principled evaluation:** the pipeline is evaluated with RAGAS (faithfulness, answer relevance, context precision/recall) rather than anecdotal testing, making the results reproducible and comparable.

6. **Intent-aware routing via fine-tuned DistilBERT:** a DistilBERT model fine-tuned on labelled support intents routes every query: predicted intents filter retrieval to the right document category, drive support analytics, and force immediate human escalation for sensitive intents (e.g., legal complaints) — demonstrating transfer learning with transformers as an integral pipeline stage rather than a bolt-on.

7. **Open and low-cost by design:** the entire stack uses open-source components (sentence-transformers, FAISS/ChromaDB, open LLMs), making enterprise-grade grounded support affordable for small businesses — directly serving the SDG mapping above.